

BBO capstone project presentation template

Develop a concise presentation that captures:

1. An overview of your BBO approach

- Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
- Briefly describe your overall strategy and how your process fits together.

The core objective of my BBO capstone project is to maximise the output of eight unknown functions using only a small number of sequential queries. Because the true equations are hidden, the task is less about solving a known formula and more about learning from feedback, updating beliefs, and choosing the next point carefully.

My overall strategy has been iterative. I began with broad exploration to cover different regions of each search space, then shifted toward local refinement once repeated patterns started to appear. In practice, the process combines simple heuristics, trend tracking, and surrogate-model thinking: compare returned outputs, identify stronger regions, adjust step size, and decide whether to continue, refine, or pull back. The process fits together as a loop of query, observe, interpret, and update.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

My strategy changed significantly after the early rounds. At first, I made larger exploratory jumps because I had very little evidence. As more query-output pairs accumulated, I stopped treating every new point as equally informative and began to prioritise repeated local evidence. Later rounds used smaller, more conservative moves around better-performing regions instead of broad jumps across the whole space.

These changes were influenced mainly by data trends and model performance rather than intuition alone. When a function improved consistently, I continued in that direction. When aggressive extrapolation reduced performance, I moved back toward a prior stronger region. The heuristics that now guide my decisions are: trust repeated improvements more than one-off spikes, use small refinements near promising areas, and treat unstable functions as candidates for cautious re-exploration rather than blind exploitation.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

The strongest trends in my data came from Functions 5, 7, and 8, where repeated local adjustments produced clear gains over time. Function 5 showed the clearest directional pattern: keeping the first coordinate near zero and the second near one while gradually decreasing the third coordinate and increasing the fourth produced a strong sequence of improvements. Functions 7 and 8 also developed compact high-performing neighborhoods, suggesting that some search spaces contain stable local basins rather than isolated lucky points.

Other functions behaved differently. Function 4 improved more slowly and seemed to benefit from returning toward an earlier strong region. Function 2 peaked early and later local moves did not consistently outperform that first strong area. Function 1 stayed near zero across many rounds, which suggests either a very flat surface or an objective whose useful structure is hard to detect with limited sampling. These observations changed my understanding of the search process: not all functions reward the same level of exploitation, and some require more patience or broader coverage than others.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

I balance exploration and exploitation by using the observed stability of each function as the main signal. When several nearby points improve together, I exploit with small step sizes. When the pattern is inconsistent or performance worsens, I switch back toward exploration or retreat to a previously stronger region. This helps me avoid overcommitting to one noisy result while still using the information I have already paid to collect.

Two examples illustrate this. First, Function 5 benefited from steady exploitation, and repeated small moves in the same direction continued to raise the output substantially. Second, Functions 2 and 6 showed that aggressive moves away from stronger earlier regions could hurt performance, so I adjusted by shrinking the step size and moving closer to the local cluster of better points. When results do not match expectations, I handle uncertainty by re-checking the recent history, reducing the size of the next move, and treating surprising results as signals to test the local neighborhood rather than assuming the trend will continue unchanged.

5. Next steps and reflection

- Outline your planned actions or experiments to improve the model's performance.
- Include two or three sentences connecting your work to the broader ML landscape.
- Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

My next steps are to make the optimisation process more systematic without losing transparency. I would like to organise the data by function, plot the query trajectories, and test a lightweight surrogate such as Gaussian process regression or local regression to compare against my current heuristic approach. I would also separate observed data from proposed-but-not-yet-evaluated queries more carefully so the decision trail is easier to reproduce.

This work connects directly to the broader ML landscape because it mirrors how real optimisation problems are handled in practice: with limited budgets, incomplete knowledge, and constant trade-offs between exploration, exploitation, and model complexity. It also shows why transparency matters in ML; a method is more useful when others can understand how decisions were made and where the approach may fail. If I were communicating these results to a non-technical stakeholder, I would say that the project is essentially a disciplined search for better settings: we test small changes, keep the adjustments that improve performance, and back away when the evidence suggests a region is becoming less useful.